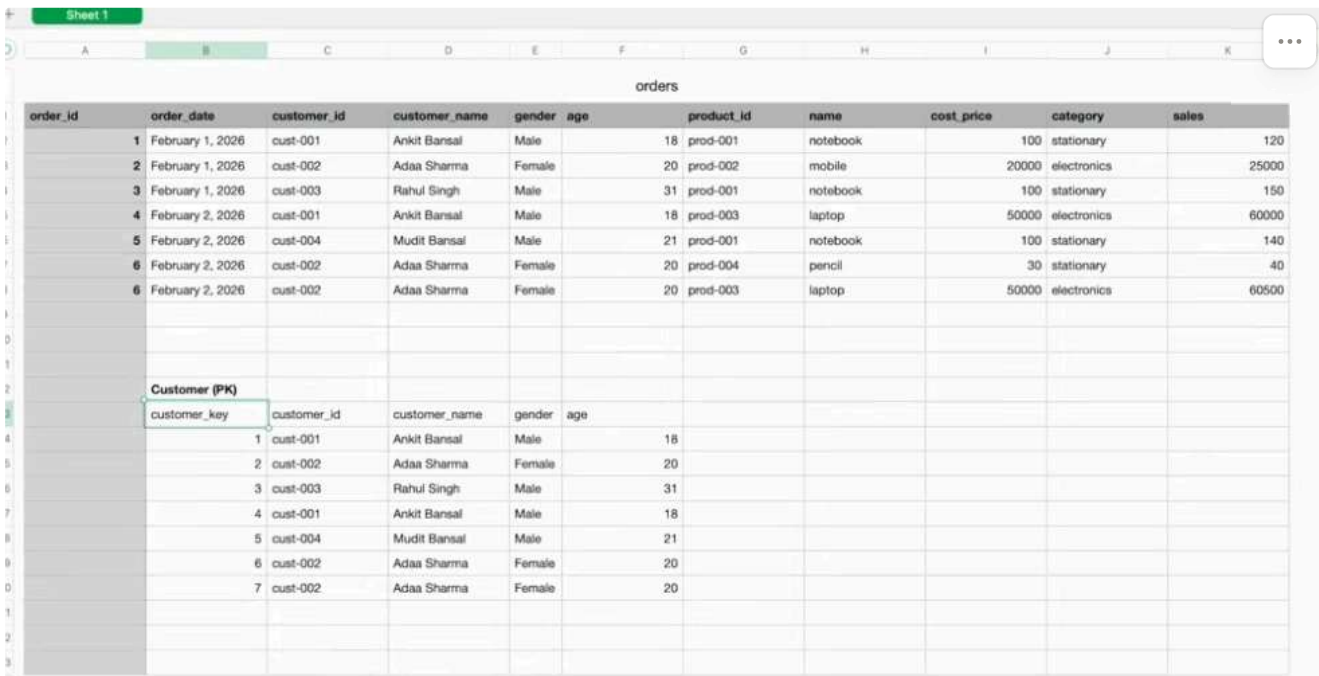
datawarehousing



orders

order_id	order_date	customer_id	customer_name	gender	age	product_id	name	cost_price	category	sales
1	February 1, 2026	cust-001	Ankit Bansal	Male	18	prod-001	notebook	100	stationary	120
2	February 1, 2026	cust-002	Adaa Sharma	Female	20	prod-002	mobile	20000	electronics	25000
3	February 1, 2026	cust-003	Rahul Singh	Male	31	prod-001	notebook	100	stationary	150
4	February 2, 2026	cust-001	Ankit Bansal	Male	18	prod-003	laptop	50000	electronics	60000
5	February 2, 2026	cust-004	Mudit Bansal	Male	21	prod-001	notebook	100	stationary	140
6	February 2, 2026	cust-002	Adaa Sharma	Female	20	prod-004	pencil	30	stationary	40
6	February 2, 2026	cust-002	Adaa Sharma	Female	20	prod-003	laptop	50000	electronics	60500

Customer (PK)

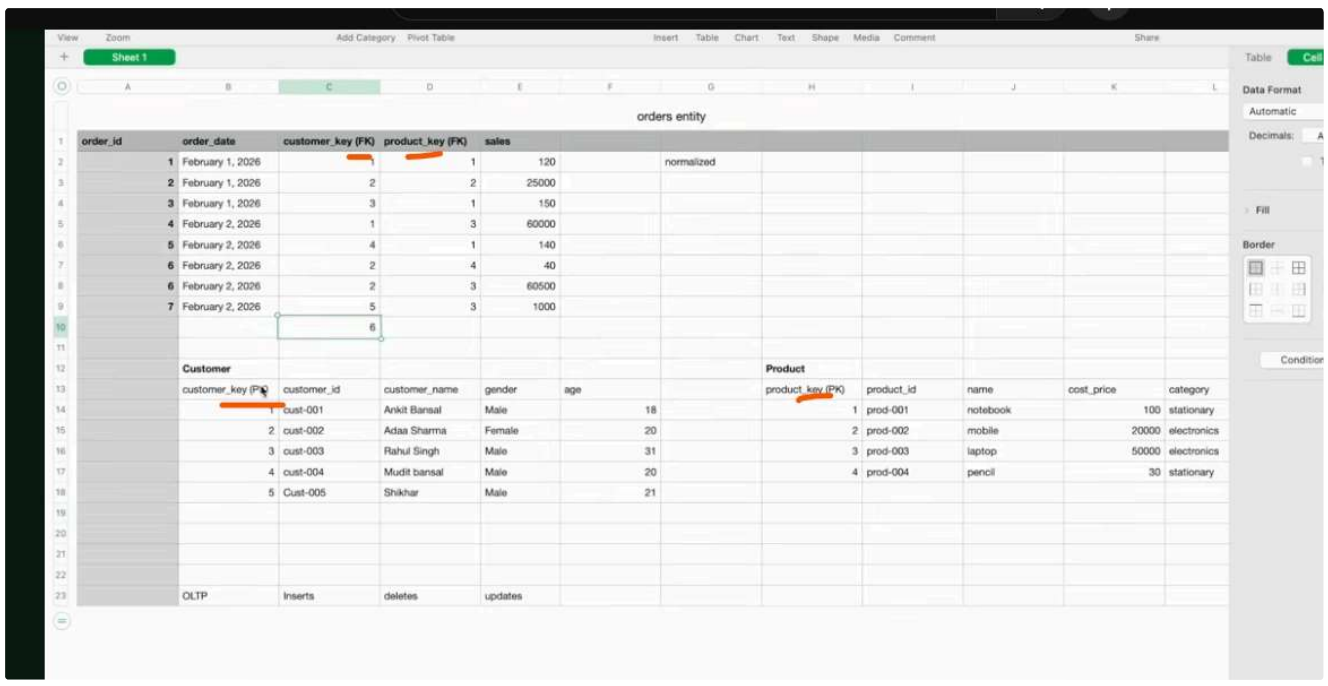
customer_key	customer_id	customer_name	gender	age
1	cust-001	Ankit Bansal	Male	18
2	cust-002	Adaa Sharma	Female	20
3	cust-003	Rahul Singh	Male	31
4	cust-001	Ankit Bansal	Male	18
5	cust-004	Mudit Bansal	Male	21
6	cust-002	Adaa Sharma	Female	20
7	cust-002	Adaa Sharma	Female	20

PK/SK arrows (for the diagram)

- `customer_dim.customer_sk` (PK) → `fct_orders.customer_sk` (FK)
- `product_dim.product_sk` (PK) → `fct_orders.product_sk` (FK)

> Legend: **PK** (Primary Key) → **FK** (Foreign Key). **SK** = Surrogate Key (usually the PK in dimension tables).

data integrity(used constraints like PK,FK)



orders entity

order_id	order_date	customer_key (FK)	product_key (FK)	sales
1	February 1, 2026	1	1	120
2	February 1, 2026	2	2	25000
3	February 1, 2026	3	1	150
4	February 2, 2026	1	3	60000
5	February 2, 2026	4	1	140
6	February 2, 2026	2	4	40
6	February 2, 2026	2	3	60500
7	February 2, 2026	5	3	1000
		6		

normalized

Customer

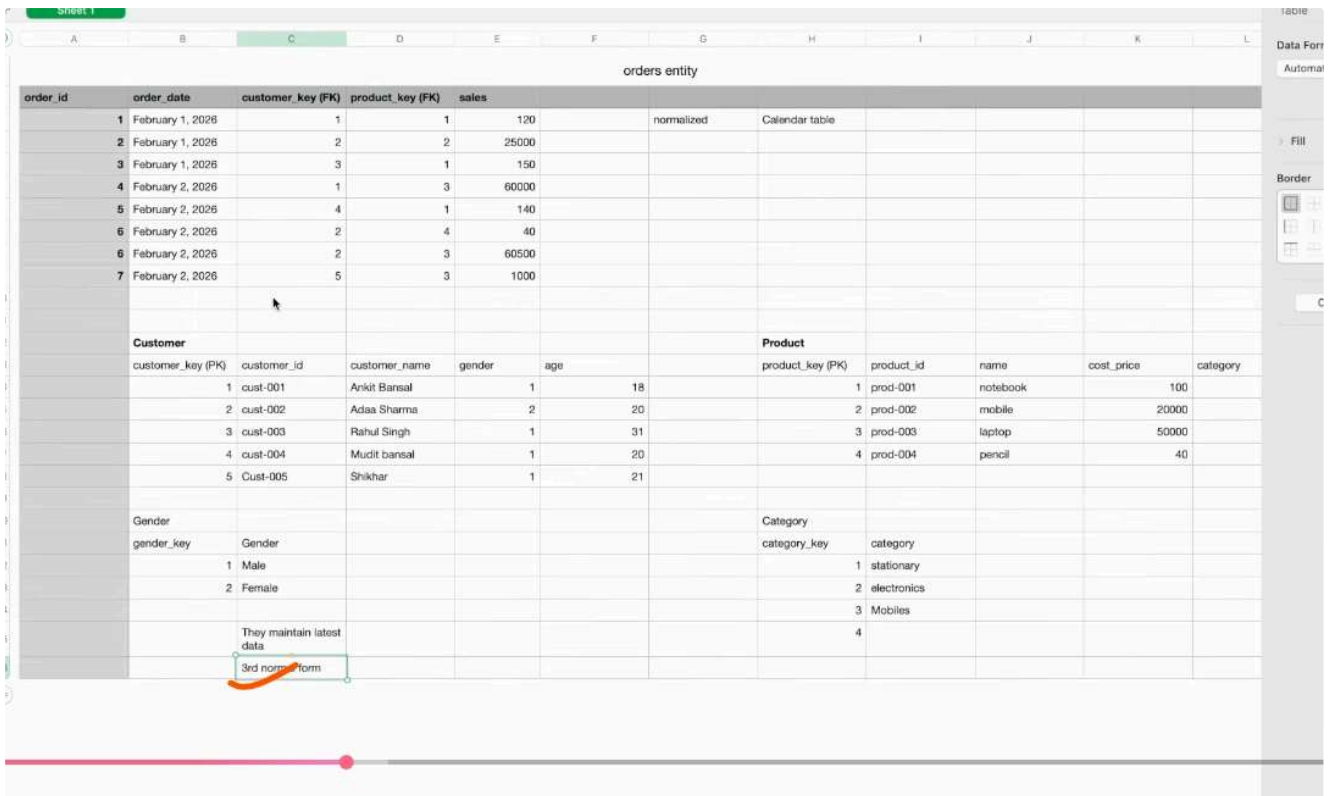
customer_key (PK)	customer_id	customer_name	gender	age
1	cust-001	Ankit Bansal	Male	18
2	cust-002	Adaa Sharma	Female	20
3	cust-003	Rahul Singh	Male	31
4	cust-004	Mudit bansal	Male	20
5	Cust-005	Shikhar	Male	21

Product

product_key (PK)	product_id	name	cost_price	category
1	prod-001	notebook	100	stationary
2	prod-002	mobile	20000	electronics
3	prod-003	laptop	50000	electronics
4	prod-004	pencil	30	stationary

OLTP	Inserts	deletes	updates

we have created seperate table for category and gneder to reduce table size:



orders entity

order_id	order_date	customer_key (FK)	product_key (FK)	sales			
1	February 1, 2026	1	1	120		normalized	Calendar table
2	February 1, 2026	2	2	25000			
3	February 1, 2026	3	1	150			
4	February 2, 2026	1	3	60000			
5	February 2, 2026	4	1	140			
6	February 2, 2026	2	4	40			
6	February 2, 2026	2	3	60500			
7	February 2, 2026	5	3	1000			

Customer

customer_key (PK)	customer_id	customer_name	gender	age
1	cust-001	Ankit Bansal	1	18
2	cust-002	Adaa Sharma	2	20
3	cust-003	Rahul Singh	1	31
4	cust-004	Mudit bansal	1	20
5	Cust-005	Shikhar	1	21

Product

product_key (PK)	product_id	name	cost_price	category
1	prod-001	notebook	100	
2	prod-002	mobile	20000	
3	prod-003	laptop	50000	
4	prod-004	pencil	40	

Gender

gender_key	Gender
1	Male
2	Female

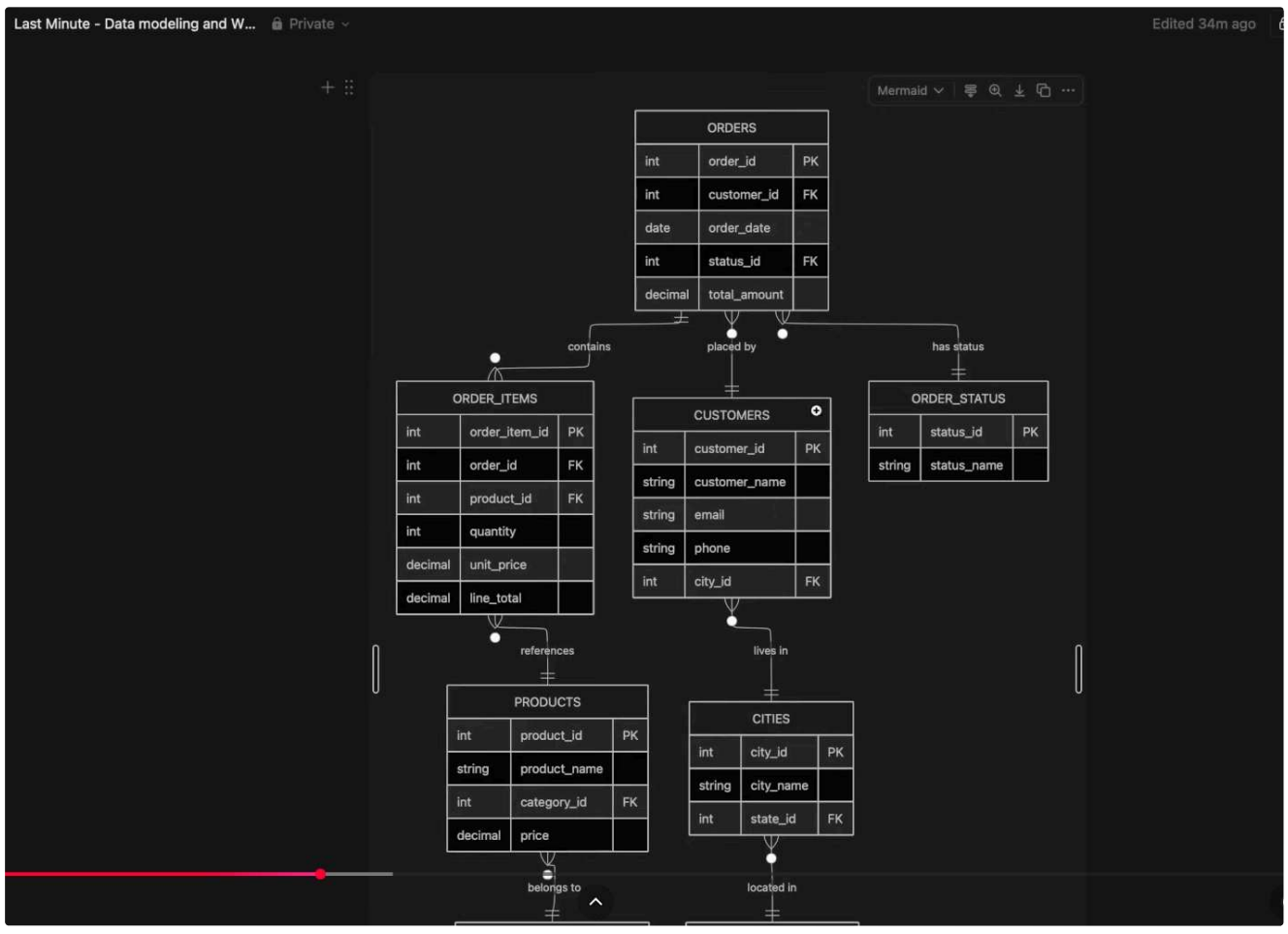
Category

category_key	category
1	stationary
2	electronics
3	Mobiles
4	

They maintain latest data

3rd normal form

OLTP system has entities orders entity customer and product entities:



OLTP maintains latest data , they dont maintain history.

OLTP—queries becomes for aggregations for analytics purpose:

OLAP uses Dimension modelling: FACTS and Dimensions

CustomerDIM amd productDIM tables i will create

M mithlesh MMS Apr 8

Add: “Facts and Dimension

3rd normal form									
			OLAP						
		fct_orders	Dimensional modelling						
		order_id	order_date	customer_sk (FK)	product_ (FK)	sales			
		1	February 1, 2026	1	1	120			
		2	February 1, 2026	2	2	25000			
		3	February 1, 2026	3	1	150			
customer_dim						product_dim			
						product_sk	product_id	name	cost_p
customer_sk	customer_id	customer_name	gender	age		1	prod-001	notebook	
1	cust-001	Ankit Bansal	Male	18		2	prod-002	mobile	
2	cust-002	Adaa Sharma	Female	20		3	prod-003	laptop	
3	cust-003	Rahul Singh	Male	31		4	prod-004	pencil	
4	cust-004	Mudit bansal	Male	20					

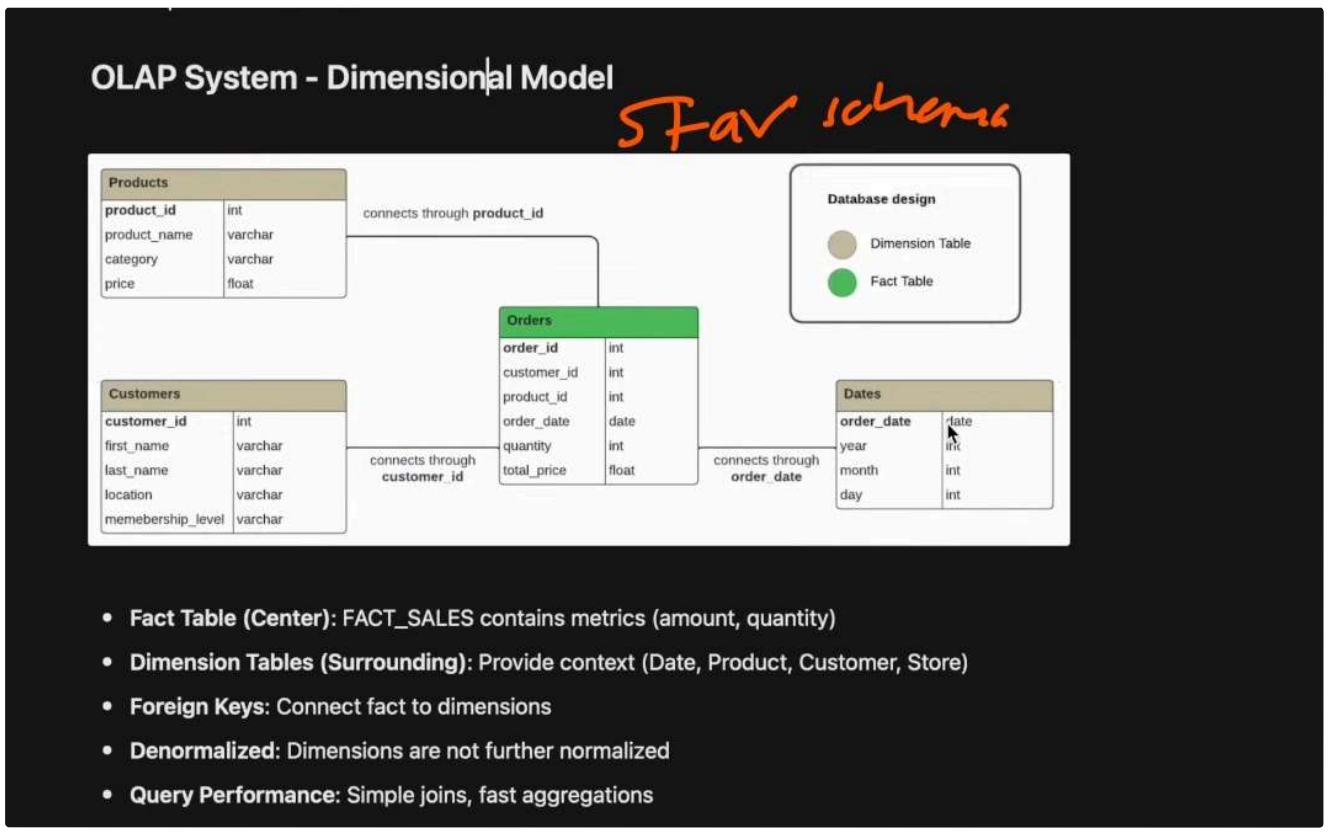
Facts and Dimension tables:

OLAP

FACT

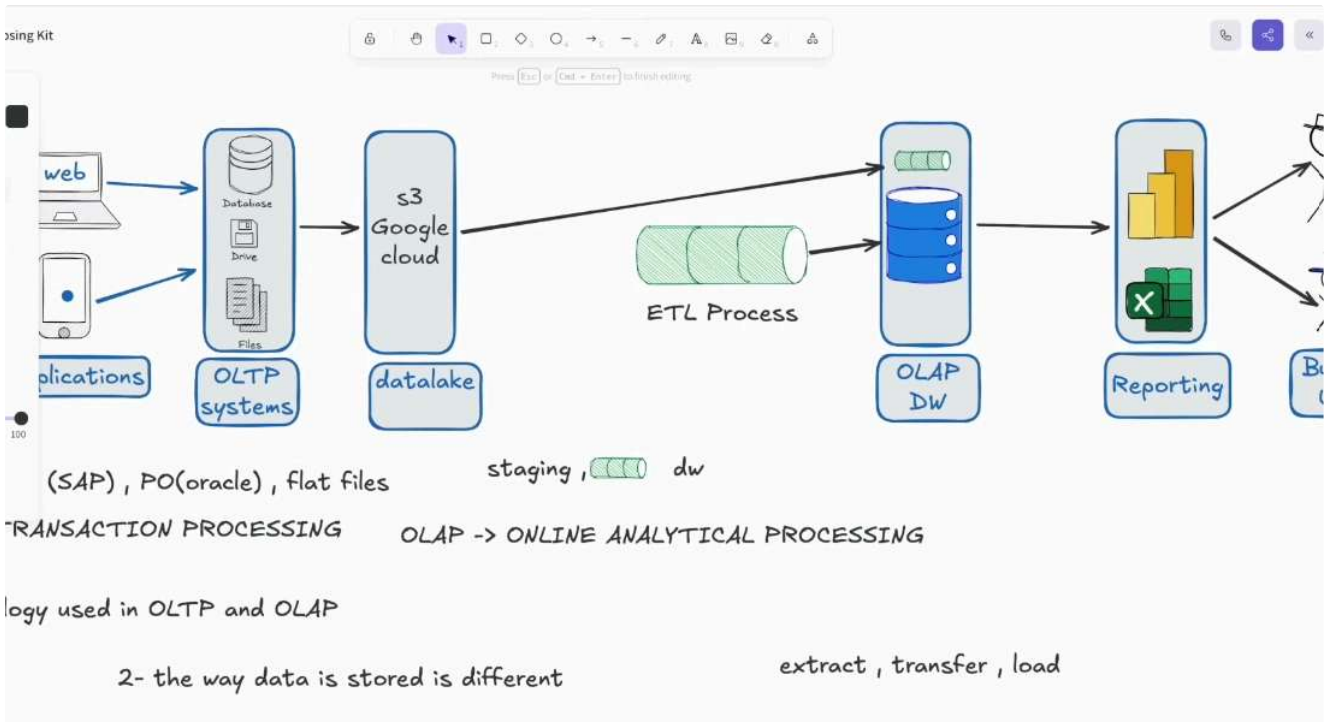
Star schema

data									
3rd normal form									
		fct_orders	OLAP Dimensional modelling						
		order_id	order_date	customer_sk (FK)	product_sk (FK)	sales			
		1	February 1, 2026	1	1	120			
		2	February 1, 2026	2	2	25000			
		3	February 1, 2026	3	1	150			
customer_dim						product_dim			
						product_sk	product_id	name	cost_price
customer_sk	customer_id	customer_name	gender	age		1	prod-001	notebook	100
1	cust-001	Ankit Bansal	Male	18		2	prod-002	mobile	20000
2	cust-002	Adaa Sharma	Female	20		3	prod-003	laptop	50000
3	cust-003	Rahul Singh	Male	31		4	prod-004	pencil	40
4	cust-004	Mudit bansal	Male	20					



Data Warehouse is designed for OLAP workloads for analystical purpose gathering data from multiple warehouses.

Time variant and nonvolatile and integrated(data comes from different sources)



OLTP	OLAP
normalized,	denormalized
short query and lonng queries	

snowflake schema — dimension tables are further devided into further sub dimensions

normalized dimensions

more joins

Fact tables:—measurable dimensional events

it contains foreign key to dimesions

OLAP

data

3rd normal form

FACT

star schema

fct_orders — OLAP Dimensional modelling

order_id	order_date	customer_sk (FK)	product_sk (FK)	sales
1	February 1, 2026	1	1	120
2	February 1, 2026	2	2	25000
3	February 1, 2026	3	1	150

customer_dim

customer_sk	customer_id	customer_name	gender	age
1	cust-001	Ankit Bansal	Male	18
2	cust-002	Adaa Sharma	Female	20
3	cust-003	Rahul Singh	Male	31
4	cust-004	Mudit bansal	Male	20

product_dim

product_sk	product_id	name	cost_price	ca
1	prod-001	notebook	100	st
2	prod-002	mobile	20000	el
3	prod-003	laptop	50000	el
4	prod-004	pencil	40	st

Factless Fact table—> no measure numeric facts it capture events,contains student attendance,promotion eligibility and websit page visits

SCD1 —> overwrite

SCD2 —> Add new row with surrogate key with effective dates

SCD1

customer_sk	customer_id	customer_name	gender	age	city	
1	cust-001	Ankit Bansal	Male	18	Mysore	
2	cust-002	Adaa Sharma	Female	20	Bangalore	
3	cust-003	Rahul Singh	Male	31	Delhi	
4	cust-004	Mudit bansal	Male	20	Mumbai	
	scd					
	Type 1					
customer_sk	customer_id	customer_name	gender	age	city	last_update_date
1	cust-001	Ankit Bansal	Male	18	Mysore	1-Jan
2	cust-002	Adaa Sharma	Female	20	Bangalore	1-Jan
3	cust-003	Rahul Singh	Male	31	Delhi	1-Jan
4	cust-004	Mudit bansal	Male	20	Mumbai	1-Jan
	10-Jan					

Table 1

customer_sk	customer_id	customer_name	gender	age	city	
1	cust-001	Ankit Bansal	Male	18	Mysore	
2	cust-002	Adaa Sharma	Female	20	Bangalore	
3	cust-003	Rahul Singh	Male	31	Delhi	
4	cust-004	Mudit bansal	Male	20	Mumbai	
	scd					
	Type 1					
customer_sk	customer_id	customer_name	gender	age	city	last_update_date
1	cust-001	Ankit Bansal	Male	18	Bangalore	10-Jan
2	cust-002	Adaa Sharma	Female	20	Bangalore	1-Jan
3	cust-003	Rahul Singh	Male	31	Delhi	1-Jan
4	cust-004	Mudit bansal	Male	20	Mumbai	1-Jan
	10-Jan					

we are overwriting no history is maintained.

in Type2 we maintain history: we have new columns effective date and end date:

	10-Jan						
	Type 2						
customer_sk	customer_id (pk)	customer_name	gender	age	city	effective_date	end_date
1	cust-001	Ankit Bansal	Male	18	Mysore	1-Jan	9-Jan
2	cust-002	Adaa Sharma	Female	20	Bangalore	1-Jan	9999-12-31
3	cust-003	Rahul Singh	Male	31	Delhi	1-Jan	9999-12-31
4	cust-004	Mudit bansal	Male	20	Mumbai	1-Jan	9999-12-31
5	cust-001	Ankit Bansal	Male	18	Bangalore	10-Jan	19-Jan
6	cust-001	Ankit Bansal	Male	18	Mumbai	20-Jan	9999-12-31
			orders				
			order_id	order_date	customer_sk		
			1	5-Jan	1		
			2	12-Jan	5		
			3	1-Feb	6		

Customer_sk 1,5,6 are surrogate keys.

Notes by Mith

Reference from Ankit B